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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Hostel Room Allocation and Student Stay Management System

Submitted by

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DECLARATION

I hereby declare that the project entitled "**Hostel Room Allocation and Student Stay Management System"** submitted in partial fulfillment of the requirements for the course **Java Programming** is a **record of my original work** carried out under the guidance of **MUGILAN R**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(PRANAV S)

CERTIFICATE

This is to certify that the project entitled **"Hostel Room Allocation and Student Stay Management System"** is a **bona fide work** carried out by **PRANAV S B.Sc.IT 'B' (2025-2028)**, **INFORMATION TECHNOLOGY**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[JAVA programming]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:COIMBATORE

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The **Hostel Room Allocation and Student Stay Management System** is a Java-based application designed to efficiently manage hostel room allocation and monitor student accommodation in educational institutions. In many colleges and hostels, room allocation and student records are handled manually, which can lead to errors, mismanagement, and lack of proper tracking.

This system automates the process of managing student details, allocating rooms based on availability, and maintaining records of student stays in an organized manner. It allows administrators to add student information, assign rooms, update occupancy status, and track room availability effectively.

The application is developed using Core Java concepts such as classes, objects, methods, loops, and exception handling. It can be implemented as a console-based application or further enhanced with a graphical user interface using Java Swing.

The main objective of this system is to improve accuracy, reduce administrative workload, and ensure efficient hostel management. This project demonstrates how Java can be applied to develop practical solutions for real-world institutional management systems.

INTRODUCTION

In today's educational institutions, efficient hostel management plays a crucial role in ensuring proper accommodation and comfort for students. Traditionally, many hostels rely on manual methods such as maintaining records on paper or using basic spreadsheets to manage room allocation and student stay details. These methods are time-consuming and prone to errors.

Manual systems often lead to issues such as incorrect room allocation, difficulty in tracking student occupancy, delays in assigning rooms, and poor record management. As the number of students increases, managing hostel operations manually becomes more complex and inefficient.

With the advancement of computer technology, automated systems have become essential for improving efficiency and accuracy in hostel management.

A computerized system helps manage student details, allocate rooms effectively, and track occupancy status in real time. It also ensures that records are maintained properly and can be accessed easily.

The **Hostel Room Allocation and Student Stay Management System** is a Java-based application designed to automate the process of room allocation and student accommodation management. The system allows users to store student information, assign rooms, monitor occupancy, and generate summaries of student stay details.

This project is developed using Core Java programming concepts such as classes, objects, methods, loops, and conditional statements. The application can be implemented as a console-based program or enhanced using Java Swing for a graphical user interface.

The main objective of this system is to reduce manual effort, improve accuracy, and provide a faster and more efficient way of managing hostel operations. This project also demonstrates how Java can be used to develop practical real-world applications.

1.1 Objective

The main objective of the **Hostel Room Allocation and Student Stay Management System** is to develop a simple and efficient application using Core Java that automates room allocation and student stay management.

Specific Objectives:

- To design and develop a Java-based system for managing student details and room allocation
- To automate the process of assigning rooms and tracking occupancy
- To reduce manual errors in record management
- To provide a user-friendly system for hostel administration
- To maintain proper records of students and room details

1.2 Overview

The Hostel Room Allocation and Student Stay Management System is a menu-driven application developed using Java.

The system allows the user to:

- Store student information
- Allocate rooms based on availability
- Track room occupancy
- Manage student stay details

The system simplifies hostel management by eliminating manual work and reducing errors. It maintains records in an organized manner and ensures efficient room utilization.

The application can be developed as a console-based program or enhanced with Java Swing.

1.3 Features

- **Student Details Management** – Store and manage student information
- **Room Allocation** – Assign rooms to students based on availability
- **Room Availability Tracking** – Monitor occupied and vacant rooms
- **Stay Management** – Track student stay duration
- **Record Maintenance** – Maintain proper hostel records
- **User-Friendly Interface** – Simple menu-driven system
- **Fast Processing** – Saves time and reduces manual work

PROGRAM ANALYSIS

2.1 Problem Definition

In many hostels, room allocation and student management are handled manually.

Common Problems:

- Difficulty in tracking room availability
- Errors in room allocation
- Time-consuming record maintenance
- Difficulty in managing large number of students
- Lack of proper documentation
- Inefficient hostel management

Hence, an automated system is required to manage hostel operations efficiently.

2.2 Project Overview

The system is a Java-based application that simplifies hostel room allocation and student stay management.

Functionalities:

- Adding student details
- Allocating rooms
- Tracking room occupancy
- Updating student stay details
- Generating summaries

2.3 Scope and Limitations

Scope:

- Managing student information
- Allocating hostel rooms
- Tracking room availability
- Maintaining records

Applications:

- College hostels
- University accommodations
- Paying guest systems

SYSTEM DESIGN

3.1 Modules

1. Student Management Module

Stores student details

Example:

- Arjun
- Rahul
- Priya

---

2. Room Allocation Module

Assigns rooms to students

Functions:

- Allocate room number
- Check availability

---

3. Stay Management Module

Tracks student stay

Functions:

- Entry date
- Exit date

---

4. Room Tracking Module

Monitors occupancy

- Available rooms
- Occupied rooms

---

5. Summary Module

Displays final details

Example Output:

------ HOSTEL SUMMARY ------

Student: Arjun

Room No: 101

Status: Occupied

---

6. Exit Module

Safely exits the system

METHODOLOGY

Development Stages:

- Problem Analysis
- System Design
- Program Development
- Testing
- Documentation

4.6 Algorithm

Start
Enter student details
Assign room
Check availability
Update occupancy
Display summary
End

4.7 Flowchart

Start → Enter Student → Allocate Room → Check Availability → Display Summary → End

---

SOURCE CODE (MODIFIED)

```
import java.util.Scanner;

class HostelSystem {

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

String studentName;
int roomNumber;
boolean isOccupied = false;

int choice;
```

```
System.out.println("===== Hostel Room Allocation System =====");

// Student Details
System.out.print("Enter Student Name: ");
studentName = sc.nextLine();

System.out.println("Student " + studentName + " added successfully.\n");

do {
// Menu
System.out.println("\n--- MENU ---");
System.out.println("1. Allocate Room");
System.out.println("2. View Room Status");
System.out.println("3. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();

switch (choice) {

case 1:
// Room Allocation
if (!isOccupied) {
System.out.print("Enter Room Number: ");
roomNumber = sc.nextInt();
isOccupied = true;
```

```
System.out.println("Room " + roomNumber + " allocated
successfully!");
} else {
System.out.println("Room already occupied!");
}
break;

case 2:
// Room Status
if (isOccupied) {
System.out.println("Room is occupied by " + studentName);
} else {
System.out.println("Room is available.");
}
break;

case 3:
System.out.println("Exiting system... Thank you!");
break;

default:
System.out.println("Invalid choice! Try again.");
}

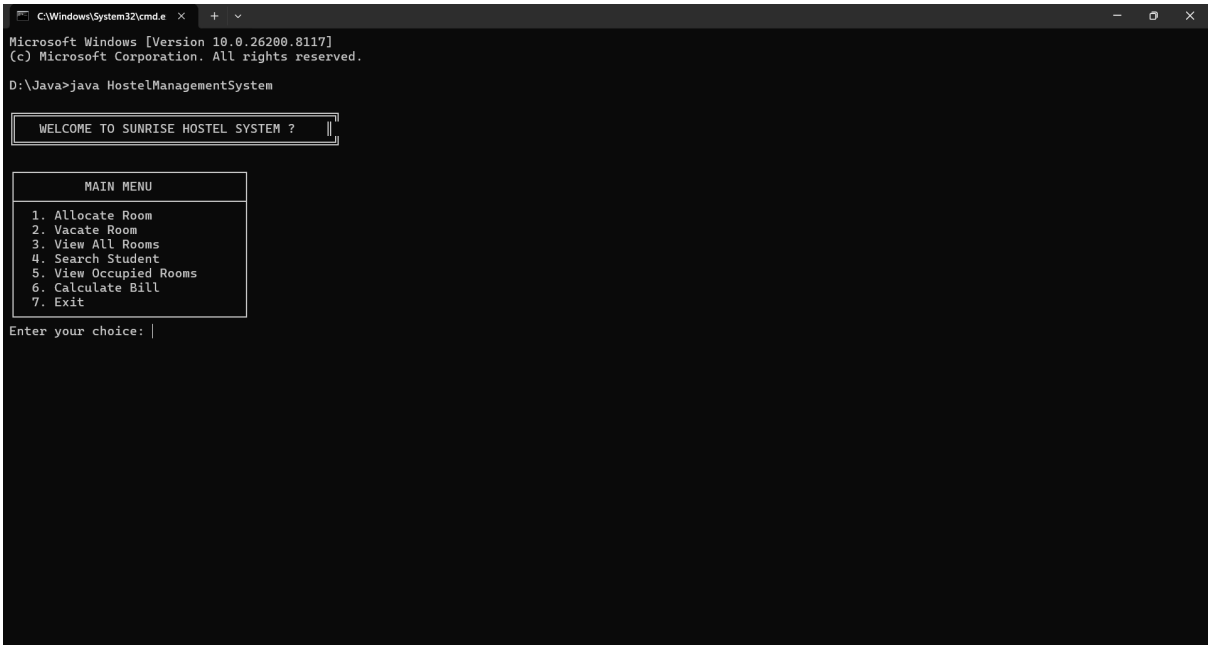
} while (choice != 3);
```

```
sc.close();
    }
}
```

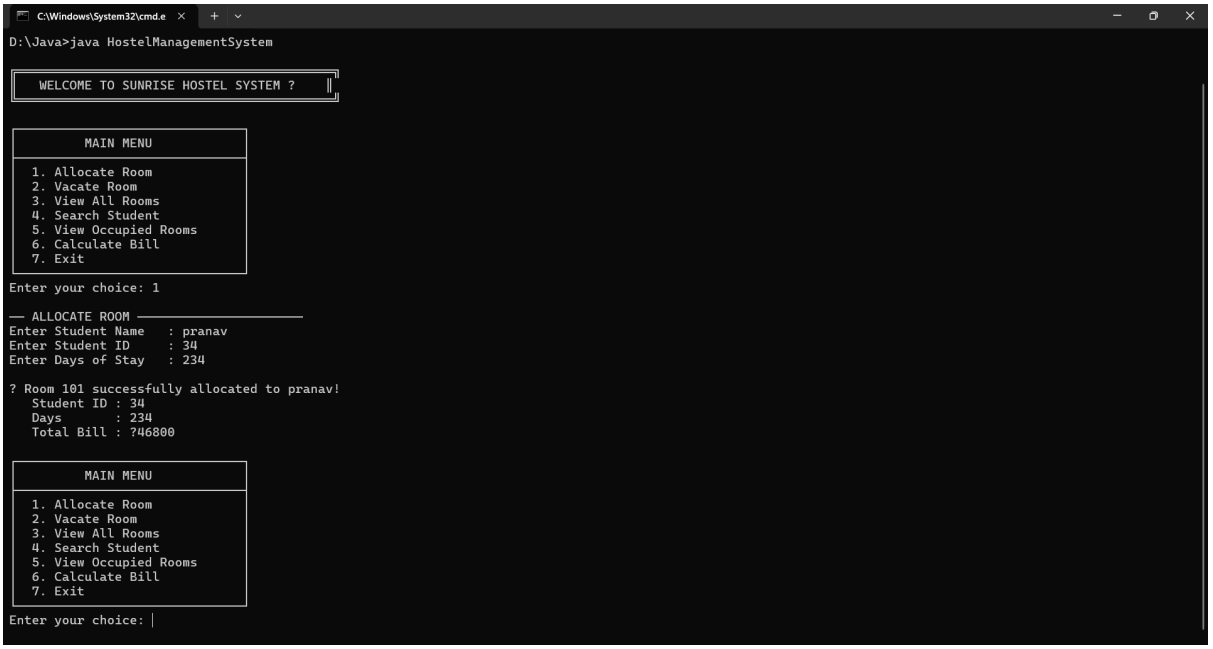
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1. INPUT AND OUTPUT SCREENS

INPUT_SCREEN:



OUTPUT SCREEN:



3. DISCUSSION & CONCLUSION

7.1 Discussion

The Hostel Room Allocation and Student Stay Management System was developed to automate the process of managing student details, allocating hostel rooms, and tracking student stay efficiently. The system allows users

to store student information, assign rooms based on availability, and generate summaries of room allocation and occupancy status.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, and object-oriented principles were applied. The system successfully performs room allocation and stay management with proper tracking of room occupancy.

The system was tested with different inputs to ensure that it performs the required operations correctly. The results show that the system reduces manual effort and improves the efficiency of hostel management. It also minimizes errors in room allocation and helps maintain accurate records of students and rooms.

7.2 Conclusion

The Hostel Room Allocation and Student Stay Management System was developed to provide an efficient solution for managing hostel operations. The system enables administrators to store student details, allocate rooms, and monitor occupancy in an organized manner.

Core Java concepts such as loops, conditional statements, methods, and object-oriented programming were effectively used in developing the system. The application performs all required functions accurately and efficiently.

The system was tested with various inputs and produced correct results, ensuring reliability. Overall, the project reduces manual workload, improves accuracy, and provides a simple and effective way to manage hostel room allocation and student stay records.

3.REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com